CHAPTER 6

Analog In, Analog Out

The Arduino interface board, just like most other computers, is a digital system that operates on binary values: 1 and 0, on and off, true and false. What if we wanted to know not just whether it was hot or cold outside but rather *how* hot or cold it was? Or instead of turning an LED on or off, maybe we want to dim or even fade an LED? To do this we need to either convert an analog reading into something that the Arduino can understand or, vice versa, approximate an analog output using digital signals. Where a digital signal has two possible states, an analog signal has many. Because we are dealing with electrical signals, digital states on the Arduino are usually considered to be 0v for `LOW` or off and +5v for `HIGH` or on. Analog signals on the other hand might represent any voltage between 0 and +5 volts, infinitely more than just on and off.

This chapter will specifically address the available hardware and the functions that we need to use them, in order to read, interpret, and output analog signals. We will discuss how a digital microcontroller can use both analog and pseudo-analog signals—pseudo-analog signals are signals simulated by digital means—converting and manipulating these signals as needed. We will look at a project called Telematic Breath that takes advantage of both reading and writing analog signals, followed by a more in-depth discussion of the various analog functions and how they work. We will also look at a way to "see" the analog input as the Arduino sees it, wrapping up the chapter with methods for mapping these values to more usable forms.

What's needed for this chapter:

- Arduino Uno
- Modern Device MD0550 wind sensor
- 2N2222 or 2N3904 NPN transistor or similar
- 5 volt DC brushless fan 50mm × 50mm or similar
- 10 kilohm CdS photocell or similar
- 10 kilohm and 1 kilohm ¼ watt resistors
- Hookup wires
- Solderless breadboard